Course Outline

Overview

In this present highly competitive and global environment, any professional and progressive organization that aims to survive, grow significantly in the marketplace and then sustain the growth must unshackle itself from ad-hoc ways of looking at data. Instead, it should use analytic tools for converting data into information for effective decisions and strategy formulation.

An 'end-to-end' Analytics problem involves multiple complex steps. One needs to define the scope of the problem, prepare the data, identify the appropriate tools and techniques and finally present the 'solution' in a way that is actionable. This art of modeling a business problem, then finding the solution using analytics, and finally implementing it successfully are some key tenets of how learning is delivered in the DSML program.

Welcome to the **Pre-Work** course of the program. In this course, we will aim to introduce you to the world of python and the basics of Statistics & Probability. It is highly recommended to go through the Pre-Work if you are new to Python programming. This course will help you prepare to take on the upcoming courses.

This course starts with the landscape of Artificial Intelligence, and how AI can be applied in organizations to solve business problems and then you will learn the different Python concepts through the videos.

Python is an easy to learn, readable, dynamic, flexible, fast, and powerful tool. First we will start with the basics of Python Programming language that will provide you with the skills and knowledge required to write simple Python programs. You will familiarize with key Python concepts and this will make you comfortable with the basic Python syntax.

We will end the pre-work with a basic understanding of Statistics and Probability.

This course aims to set a strong foundation for this journey of Data Science and Machine Learning.

Course Objectives

After completing this course, you will be able to:

- Get a good appreciation of what Data Science means and what are some key aspects of it
- Understand the importance of data identification, gathering, and processing
- Understand why we need the tools and techniques that we are using
- Appreciate the importance of programming

Topics Covered

Module	Name of the topic
Introduction to Python	• Introduction to Google Colab • Variables • Data Structures • Conditional Statements • Looping Statements • Functions
Descriptive Statistics	• Basics of Statistics • Measures of Central Tendency • Measures of Dispersion • Covariance and Correlation Coefficient • Exploratory Data Analysis (EDA)

Learning Material

Module no.	Module	No. of Videos	Total Duration	No. of Test Your Understanding Quizzes
1	**Introduction to Python**	9	~1.26 hours	9
2	**Statistics and Probability**	21	~3.5 hours	21

We hope that this content will help you start nicely with the Python programming and Data Science constructs. We recommend you go through all the videos and attempt all the practice quizzes and assignments to make the most out of this course.

Practice Quiz

A practice quiz is to be submitted at the end of both the modules. The quiz contains questions from the concepts covered in the video content.

Note: This is an optional course and there are no graded quizzes/assignments in this course.

We wish you a great learning experience.

Happy Learning!

Next >

Proprietary content.©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

© 2023 All rights reserved.